

The Local Economic Impact of Data Centers

County-level measures in the contiguous United States

Ian Walsh & Logan Rosell

DATA 510 Data Science Capstone · Summer 2026

Project Overview

How does the deployment of a new data center relate to local economic conditions, including employment, income, environmental impact, and business development patterns at the county level?

- Data centers are associated with measurable changes in county-level economic and air quality indicators
- We analyze county data on income, employment, and PM2.5 to model economic impact
- Data Center count (≥ 3 per county): Air quality improves, personal income decreases
- Regional data center county (≥ 112 within 50 miles): Decrease in unemployment rate

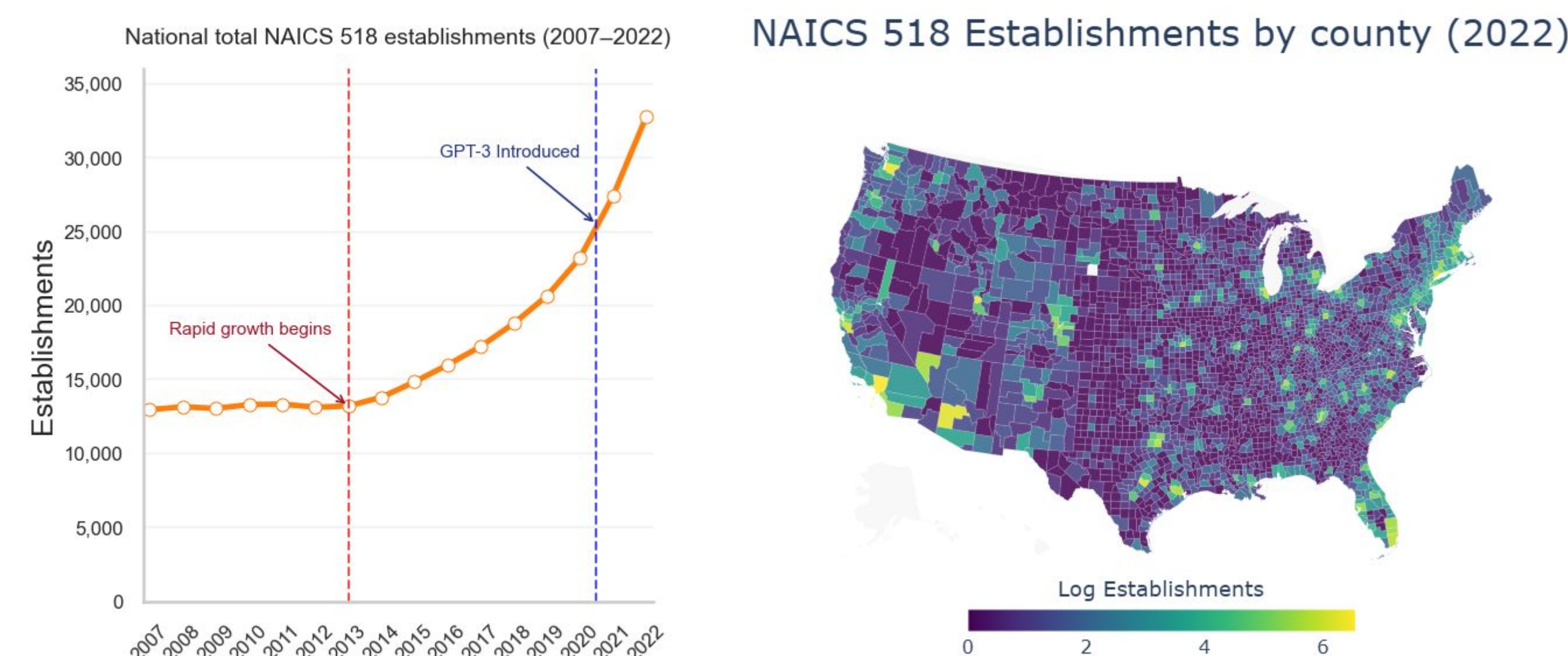


Introduction & Motivation

Data Center definition (NAICS 518): "The Computing Infrastructure Providers Data Processing Web Hosting and Related Services subsector groups establishments that provide computing infrastructure data processing services Web hosting services (except software publishing) and related services including streaming support services (except streaming distribution services)."

Why this matters:

- Data center construction has exploded since 2013
- Data centers tend to congregate concentrating infrastructure, economic activity, and environmental impact in specific counties



Impact & stakeholders:

- Residents** may see changes in jobs, wages, and environmental quality
- Communities** must understand the specific risk-reward profile when shaping local policy

Background & Related Work

- Limited existing research:** Academic work on the local economic impact of data centers is still emerging
- Regional Analysis:** Analysis of Texas counties that found data center development linked to local job growth¹
- High-level local economic effects:** County-level study of US which found data centers had positive impacts on employment, total wages, business establishments, median household income, and building permits. Gains in metropolitan counties, negligible effects in rural counties²
- Environmental externalities:** Large-scale power plant utilization and on-site diesel generators impact air quality around data centers³
- Valuing air quality impacts:** Changes in PM2.5 levels have a measurable impact on productivity⁴
- Our contribution:**
 - Incorporate spatial relationships between neighboring counties
 - Include sector-specific employment and economic ramifications of air quality impacts
 - Utilize modern machine learning techniques

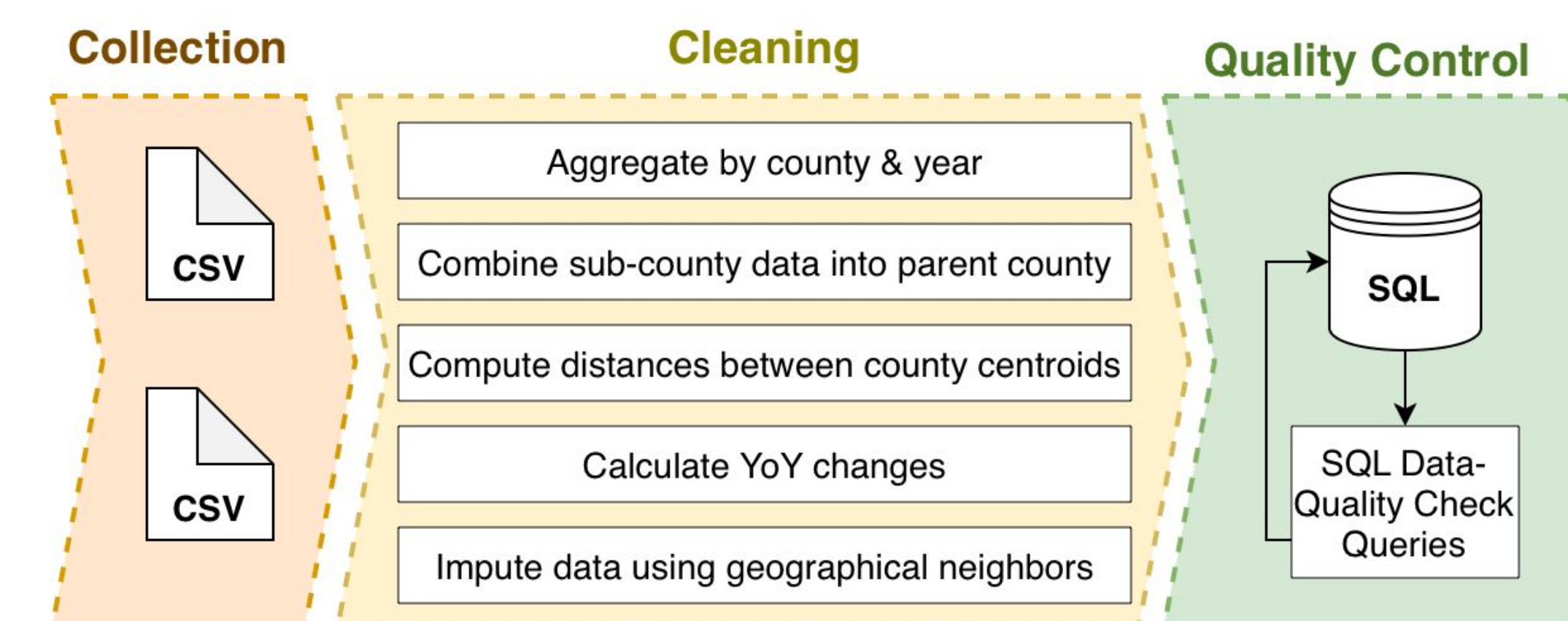
Data, Engineering & Ethics

Data & sources:

- County-level data for the contiguous U.S. between 2007 & 2022
- Data sourced from EPA, BEA, BLS, Census, and CDC

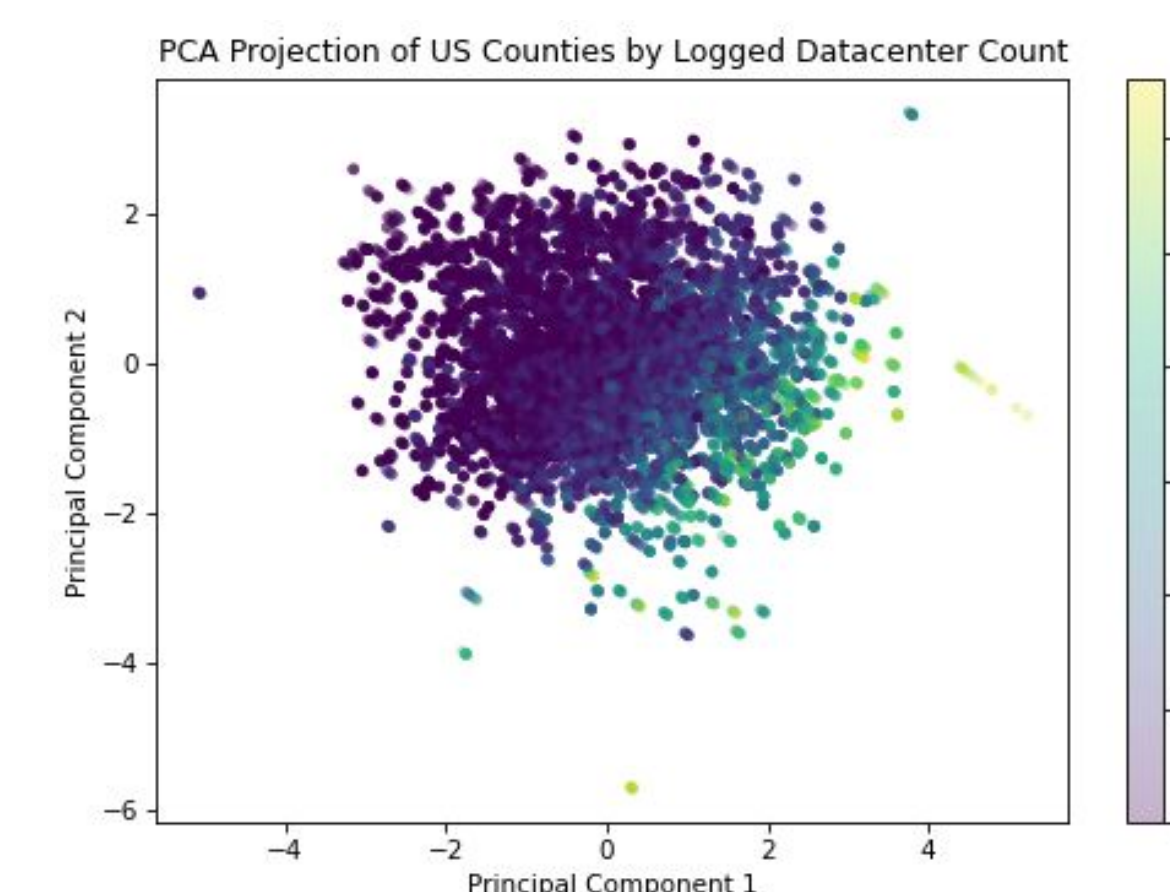
Ethics & responsible use:

- All data aggregated on county rendering no personally identifiable information
- Economic indicators may reflect demographic changes that both attract data centers and affect outcomes, introducing potential bias
- Results interpreted cautiously avoiding overstating or trivializing effects recognizing metrics like unemployment and income represent real lives

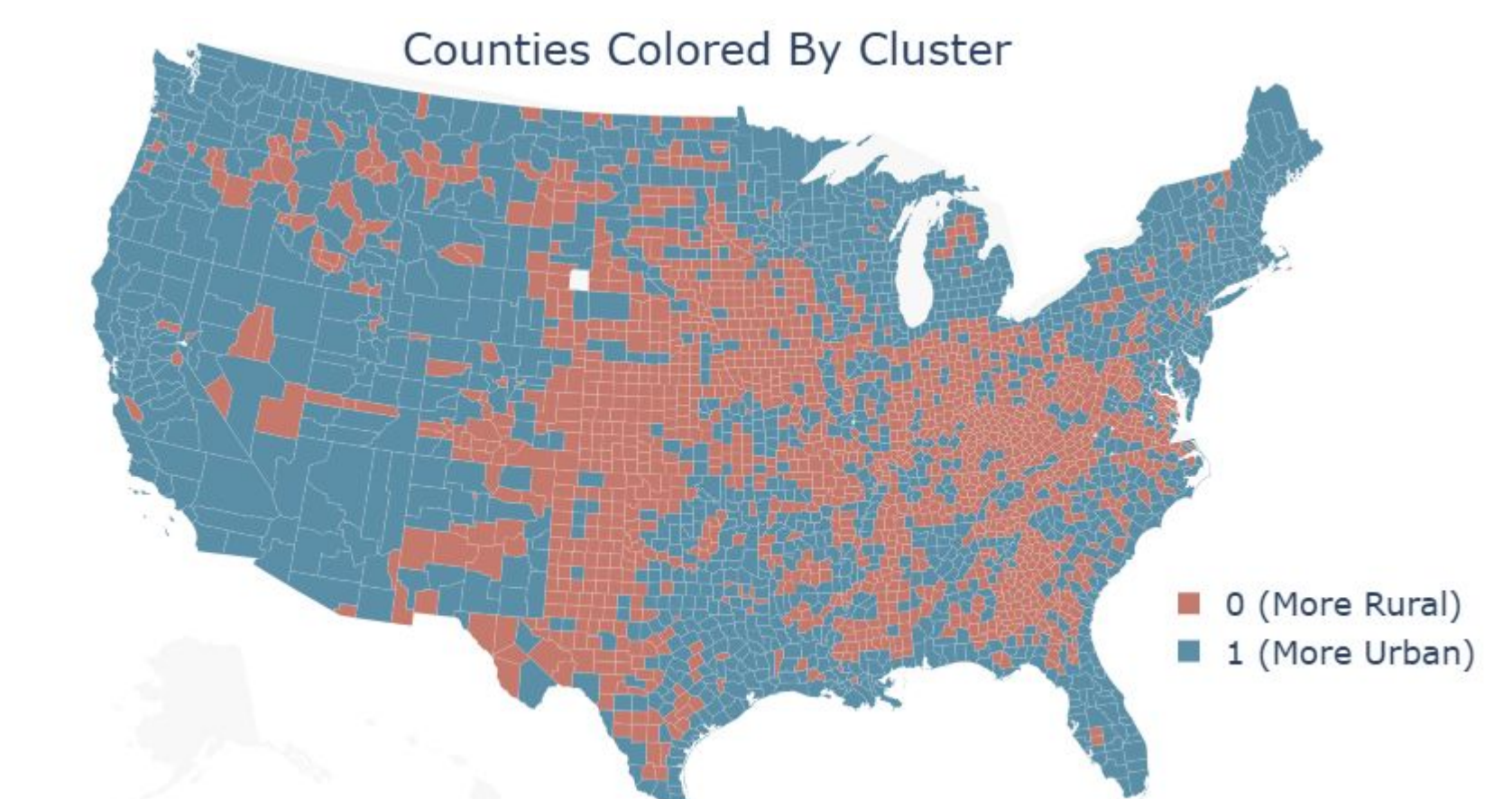


Methods & Evaluation

- PCA Projection:**
 - Used in k-means clustering
 - Smooth interpolation of log data center count along diagonal
 - No data center features included in PCA training
- Feature engineering/selection:**
 - Spatial (weighted sum/average)
 - Temporal (year-over-year difference) values
 - Correlation plots
 - Distributions by presence of data centers
 - K-means clustering
- Difference-in-difference regressions:** Compares changes over time between counties which did and did not receive new data centers
 - parallel trends assumption validates whether treated and control counties followed similar outcome trends before data center adoption
- Variational autoencoder neural network:**
 - Network of shape [128, 16, 8, 16, 128]
 - Conditional embeddings for year [16,8] and log data center count [6,4]
 - Predicted outcomes for same input sample are compared under two conditions: original case and a counterfactual case with higher logged data center count applied via decoder embedding

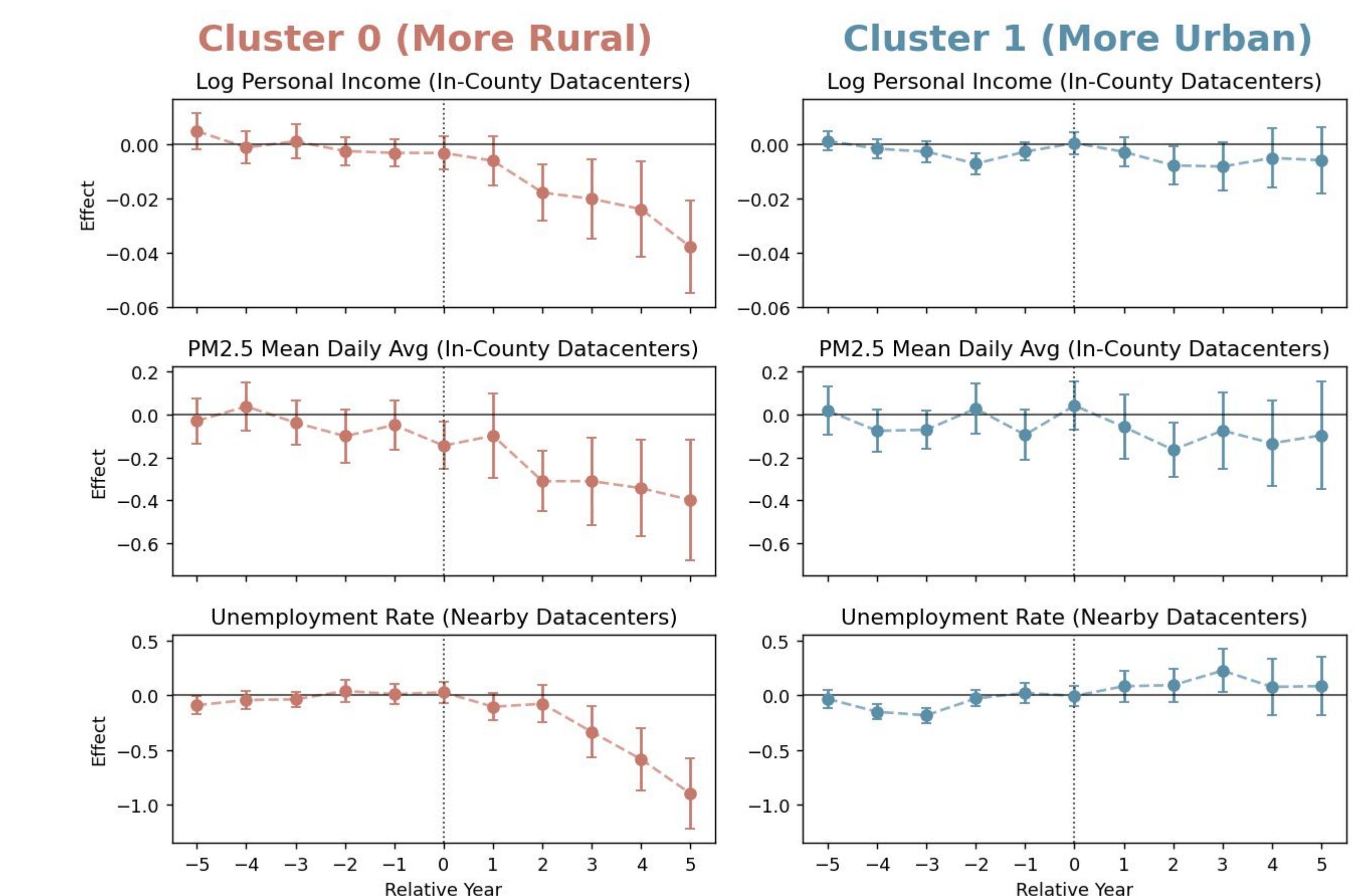


Results & Visualizations



Event-study estimates

- Additional data centers over 2 in a county
 - lower log personal income
 - lower PM2.5 levels over time
- Additional *nearby* data centers over 112 lowers unemployment over time
- Results are statistically significant in cluster 0 (rural), no significant effects in cluster 1 (urban)



Discussion, Limitations & Conclusions

Discussion & Conclusions:

- Positive impacts:** lower unemployment, higher construction activity, and lower average pollution levels
- Negative impacts:** lower personal income and lower employment in trade, transportation, industrial, and primary industries

Limitations & Future Work:

- Data center counts based on NAICS 518 establishments, likely including some non-data center facilities
- Certain spatial calculations assume even distribution of data centers within counties
- Certain spatial metrics approximate local economic conditions with inverse-distance-weighted averages or sums over the five nearest counties which may be less accurate in regions where counties are far apart or heterogeneous.

Reproducibility & References

Repository: [https://github.com/wu-msds-capstones/project-writeup-ian_and_logan] · Key dependencies: [Python 3.13, pandas, scikit-learn, seaborn, sqlalchemy]

References: [1] Michael Hick, "Data Centers and Local Job Creation", 2025. [2] Daniel Yue and Yiang Zeng, "The Local Economic Effects of Data Center Entry", 2026 [3] Nagata et. al., "The Cloud Next Door", 2025 [4] Maniloff and Schreiber, "Valuing Air Pollution's Impact on Labor Productivity in General Equilibrium", 2024 [5] <https://datacenters.atmeta.com/2025/08/kansas-city-we-are-online/> [6] https://en.wikipedia.org/wiki/375_Pearl_Street

